

CAD HACKATHON 2026

Official Rulebook

1. About the Hackathon

CAD Hackathon 2026, conducted as part of AGNI, is an open-theme engineering design competition. Participants identify a real-world engineering problem and develop a CAD-based solution. No fixed problem statement will be provided by the organizers.

2. Theme and Domains

Participants may select a problem from:

- Automotive
- Aerospace
- Agriculture
- Robotics
- Mechanical Systems
- Consumer Products

The selected project must involve a physical engineering product, component, mechanism, machine, vehicle, structure, or system in which CAD plays a significant role.

3. Eligibility

- Open to students currently enrolled in a recognized college or university.
- Students from engineering and related technical disciplines may participate.
- Inter-departmental teams are permitted.

4. Team Size

Each team shall consist of 2–4 participants.

Each participant may be part of only one team.

5. Competition Stages

- ✓ Problem Identification
- ✓ Concept Development
- ✓ CAD Development
- ✓ Final Submission
- ✓ Final Presentation and Q&A

6. Event Timeline

Time	Activity
08:00-09:00	Registration
09:00-10:00	Inauguration
10:00-10:20	Rules and briefing
10:20-10:40	Problem identification & Concept development
10:40-12:30	CAD development
12:30-01:30	Lunch
01:30-03:30	Final presentations
04:00	Prize distribution

7. Laptop and Equipment

- The organizers will NOT provide laptops or computers.
- Participants must bring their own laptops and required accessories.
- Participants are responsible for ensuring that the required CAD software is installed and functional before the event.
- Participants should carry their own chargers, adapters, mouse, and any other required accessories.

8. Problem Identification

Each team must identify and define its own engineering problem. The team should clearly establish:

- Problem
- Target user
- Existing solution
- Gap or limitation
- Design requirements
- Design constraints

9. CAD Development

Participants may use any suitable CAD software available to them, such as Creo, Siemens NX, CATIA, SolidWorks, Fusion 360, Solid Edge or another relevant CAD platform.

Depending on the project, the submission should include:

- 3D part model
- Assembly

- Engineering drawings (If possible)
- Relevant dimensions
- Material information

10. Engineering Requirements

Teams should demonstrate engineering reasoning relevant to their chosen problem. This may include.

- Material selection
- Manufacturing process
- Weight optimization
- Cost estimation
- Market impact

11. Final Presentation

Each team must present its project. Presentation time: **10 minutes**. Judge Q&A: 3 minutes.

Recommended presentation structure:

- Slide 1 – Problem Identification: problem, user, existing solution, limitation
- Slide 2 – Proposed Solution: concept, working principle, key features, sketches
- Slide 3 – CAD Design: 3D model, assembly, important components, dimensions
- Slide 4 – Engineering: material, manufacturing, calculations, cost and weight
- Slide 5 – Final Solution: final render, advantages, limitations, future improvements and application.

12. Jury Q&A

Judges may question teams on:

- Design decisions and requirements
- Material selection
- Manufacturing method
- Cost and feasibility
- Dimensions and calculations
- Failure or safety considerations

13. Judging Criteria

Criteria	Marks	Score
Problem Identification	10	
Innovation & Creativity	15	
Engineering Approach	10	
CAD Quality	15	
Functionality	10	
Manufacturing Feasibility	10	
Material & Cost Consideration	10	
Presentation	10	
Q&A	10	
TOTAL	100	

14. Submission Requirements

Each **team must submit** a folder containing:

- 01_Problem_Statement.pdf
- 02_Concept_Sketches (or) concept_images.
- 03_CAD_Model – Native CAD Files / STEP Files
- 06_Engineering_Drawings
- 07_Engineering_Report.pdf
- 08_Final_Presentation.pptx

Recommended Engineering Report length: maximum 10 pages.

15. Originality and Intellectual Property

- All submitted work must be original to the participating team.
- Downloading an existing CAD model and presenting it as original work is prohibited.
- Using another team's files or previous project work without permission is prohibited.
- Commercially available models may be referenced only when the team's own engineering development is clearly demonstrated.
- AI tools may be used for research, grammar correction, or presentation assistance, but the engineering design, CAD model, and technical decisions must be the team's own work.

16. General Rules

- Each participant can register with only **one team**.
- Teams must follow the announced start and submission times.
- Participants are responsible for saving and backing up their work.
- Teams must work within the designated event area and follow venue instructions.
- Participants must bring their own laptops and required accessories.
- Organizers may inspect submitted CAD files if originality or compliance is questioned.
- The jury's scoring decision shall be final.

17. Penalties

Violation	Penalty
Late submission	-5 marks
Exceeding presentation time	-2 marks
Missing required deliverable	-5 marks
Use of another team's work	Disqualification
Plagiarism	Disqualification
Deliberate equipment damage	Disqualification
Falsification of results	Disqualification

18. Coordinator's Note

“Great ideas mean nothing until you build them.

join now, compete hard and turn your vision into reality.”

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